

Li Shunyang

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Skills

- **Languages:** C++ (C++17/20, templates, concurrency), Python, Bash
- **Performance & Quantization:** CUDA, Triton, TensorRT, ONNX, FP8/NVFP4/W4A16 (ModelOpt, AutoRound), low-latency optimization
- **ML & Inference:** PyTorch, vLLM / vLLM-Omni, diffusion & omni-modality serving

Open-Source Contributions

vLLM-Omni Committer (52 merged PRs)

- **Quantization:** Architected the unified quantization framework (#1764) after adding online FP8 for DiTs (#1034); extended ModelOpt FP8 to Wan2.2/HunyuanVideo-1.5 (#3305) and global online FP8 to MiniMax-H3 (#5910).
- **Attention backends:** Made Blackwell a first-class diffusion target with cuDNN/FlashInfer backends and auto-routing (#3079); added TRTLLM_ATTN with Skip-Softmax (#5283) and typed SAGE FP8/INT8 attention (#5509).
- **Model integration:** Integrated HunyuanVideo-1.5 T2V/I2V (#1516), ByteDance Lance (#3710), JoyAI-VL streaming interaction serving (#4575), and LTX-2.5 T2V/I2V pipelines (#6070).
- **Cosmos3:** Fixed joint video+sound generation under Ulysses sequence parallelism (#4678) and shipped the Cosmos3-Edge recipe and correct task presets (#5313, #5596).

vLLM Contributor

- Drove early adoption of **TurboQuant** (ICLR 2026, Google Research) in core vLLM (#38280), adding a KV-cache backend with fused Triton encode/decode kernels.

Industry Experience

Developer Technology Engineer Intern, NVIDIA DevTech, Shenzhen, China

Jul 2026 – Present

- Optimized vLLM-Omni's **TensorRT-LLM attention backend** for **Blackwell SM100**, integrating generated FMHA kernels and tuning backend execution for production video-diffusion workloads.
- Optimized **Fast Ulysses** for **Blackwell SM120**, implementing a packed peer-to-peer PCIe transport and topology-aware Ulysses parallelism for multi-GPU video diffusion inference.

Backend Engineer Intern ByteDance / TikTok, Singapore

Mar 2026 – Jun 2026

- Built the end-to-end **data pipeline** for a **unified LLM-generation framework** powering **q2q (query-to-query)** and **v2q (video-to-query)** generation tasks within TikTok's recommendation architecture.
- Built scalable offline data pipelines (ingestion, cleaning, and feature processing) on the team's large-scale infrastructure to produce training data for the framework.

LLM Engineer Intern Thales DIS, Singapore

May 2025 – Dec 2025

- Scaled vLLM inference for **Qwen3-Coder-30B / Qwen3-30B-A3B-Instruct** on 2×H100 80GB GPUs, cutting total inference time for 14k samples from **26 hours to 15 minutes** ($\approx 104\times$ **speedup**).
- Fine-tuned models with **QLoRA** for code generation and completion, improving accuracy on internal JavaCard and cryptography benchmarks; co-built a dataset generator producing 10k+ training samples.

Computer Vision Lead (Extracurricular) Calibur Robotics, NUS, Singapore

Aug 2024 – Oct 2025

- Achieved **end-to-end perception latency of 2.2 ms/frame** on **NVIDIA Jetson AGX Thor** by designing a real-time pipeline spanning camera capture, **YOLOv11-Pose** detection, pose estimation, target tracking, and command output.
- Optimized the model with **ONNX** and **TensorRT** plus custom GPU kernels, reaching **1.7 ms** inference latency on Jetson AGX Thor.

Education

National University of Singapore

Aug 2023 – May 2027

- *Bachelor of Engineering in Computer Engineering (Honours), Minor in Mathematics*
- Cumulative GPA: 4.58/5.0